## APPENDIX I. DJIBOUTI: JOINT BANK-FUND DEBT SUSTAINABILITY ANALYSIS (DSA)

*The joint Bank-Fund low-income country debt sustainability analysis (LIC DSA) indicates that Djibouti's risk of debt distress continues to be high, despite some improvement in the medium-term projections. There are however important sources of vulnerability, including a worsening of financing conditions, lower than expected long-term growth rates, external shocks (such as further food and oil price increases), shortfalls in foreign direct investment, and contingent liabilities arising from government guarantees on external debt.*

### A. Background

50. **After improving considerably in 2006 following a debt restructuring agreement with Italy, Djibouti's debt situation remained broadly stable in 2007.** External debt (government and government-guaranteed) increased from 56.5 percent of GDP in 2006 to 59.5 percent of GDP in 2007, while government debt (external and domestic) declined from 63 percent to 62 percent of GDP.[6]

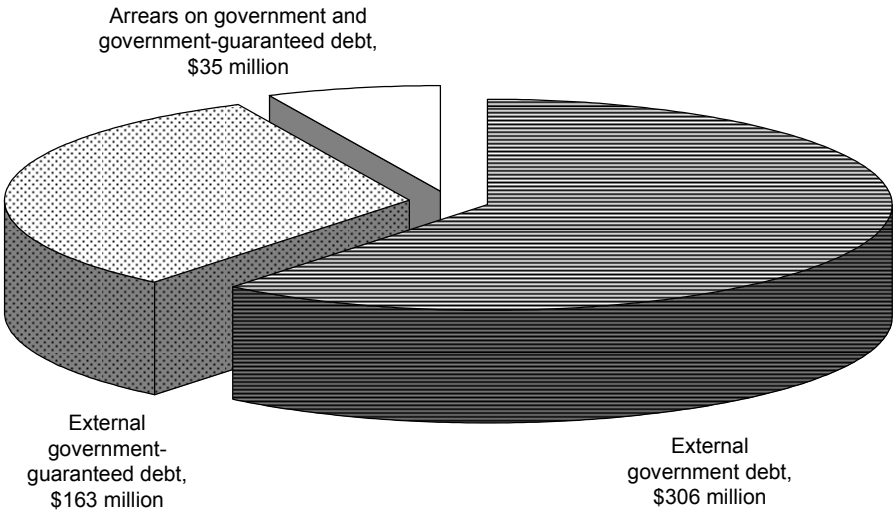
51. **Prospects for continued support from official creditors are good, and would be further enhanced under a Fund-supported program.** The country is receiving substantial financing from official multilateral and bilateral creditors to finance public investment in health, education, public housing, infrastructure, energy and water sectors, mostly in the form of earmarked project loans, including loans to state-owned enterprises under government guarantee. Major projects in the pipeline include the interconnection with the Ethiopian power grid (already under construction with financing from the African Development Bank and the European Union), a desalinization plant, the construction of new university buildings, and projects in energy, water, sanitation and social housing. Altogether, additional external lending for about $0.5 billion (equivalent to about 40 percent of average annual GDP) is projected to be disbursed between 2008 and 2013. About 40 percent of this financing had already been secured at end-2007, and the rest includes assistance pledged by Arab creditors during a Round Table held in Djibouti at end-2005. If agreement were reached on a Fund-supported program, all new lending agreements are expected to be extended on concessional terms, and additional financing could become available from the European Union, France, and other creditors in the form of budget-support loans and grants (estimated at about $40 million in 2008 and 2009). A donors' round table is scheduled for November 24–25, 2008 in Brussels, under World Bank coordination.

52. **Virtually all external debt—which amounted at end-2007 to $504 million, equivalent to 59 percent of GDP—is provided by official creditors either to the government or to state-owned enterprises under government guarantee (Appendix Table 13; Appendix Figure 3).** This includes $14 million of outstanding liabilities to the IMF, $150 million owed to the World Bank, $103 million to the African Development Bank, $57 million to the AFESD, and $51 million to bilateral Arab creditors

[6] The values for 2006 have been revised on the basis of a re-assessment of debt to Italy still outstanding after the 2006 rescheduling agreement.

(Appendix Table 14). Debt to multilateral creditors (with the notable exceptions of AFESD and OPEC countries) is on highly concessional terms, but debt to most bilateral creditors has a grant element below the 35 percent threshold considered concessional. About one-third of the debt ($163 million) is owed by state-owned enterprises under government guarantee.

**Appendix Figure 3. Djibouti: Composition of External Debt, End-2007**
(In millions of U.S. dollars)




Appendix Table 13. Amount and Composition of External and Government Debt, End-2007

| | DF(In billions) | U.S. dollars (In millions) | Percent of GDP |
|---|---|---|---|
| **External debt** | **89.5** | **503.7** | **59.3** |
| External government debt (excluding arrears) | 54.4 | 305.8 | 36.0 |
| External government-guaranteed debt | 29.0 | 163.3 | 19.2 |
| Arrears on government and government-guaranteed debt | 6.1 | 34.6 | 4.1 |
| **Government debt** | **93.8** | **527.8** | **62.2** |
| External government debt | 60.5 | 340.4 | 40.1 |
| Direct external government debt | 54.4 | 305.8 | 36.0 |
| Arrears on government and government-guaranteed external debt | 6.1 | 34.6 | 4.1 |
| Domestic government debt | 33.3 | 187.4 | 22.1 |
| Gross government credit from banks | 5.6 | 31.7 | 3.7 |
| Government debt toward the port | 0.6 | 3.4 | 0.4 |
| Domestic arrears of the government | 27.1 | 152.4 | 17.9 |

Appendix Table 14. Composition of Djibouti's External Debt (Excluding Arrears) at End-2007, by Creditor

| Creditor | Outstanding Amount (In U.S. dollars) | Outstanding Amount (In Percent of Total External Debt) |
|---|---|---|
| World Bank (IDA) | 149.8 | 31.9 |
| African Development Bank | 103.2 | 22.0 |
| AFESD | 57.1 | 12.2 |
| Islamic Development Bank | 15.7 | 3.3 |
| IMF | 14.4 | 3.1 |
| OPEC | 12.6 | 2.7 |
| IFAD | 2.7 | 0.6 |
| **Total multilateral creditors** | **355.4** | **75.8** |
| Italy | 21.3 | 4.5 |
| Kuwait | 25.2 | 5.4 |
| China | 21.3 | 4.5 |
| Saudi Arabia | 16.5 | 3.5 |
| Abu Dhabi | 9.8 | 2.1 |
| Other bilateral creditors | 19.6 | 4.2 |
| **Total external debt** | **469.1** | **100.0** |

53. **There are outstanding arrears toward external and domestic creditors but, except for arrears to Paris Club creditors, they mainly reflect weaknesses in liquidity management and do not stem from a lack of resources to service debt obligations.** Arrears on external debt service amounted at end-2007 to $35 million, equivalent to 4 percent of GDP and 7 percent of total external debt; these included about $29 million of arrears to Paris Club creditors on debt agreed before the cutoff date of March 31, 1998, which is potentially eligible for rescheduling and restructuring. Domestic arrears owed to state-owned enterprises, social security funds, public employees and government suppliers at end-2007 amounted to $152 million (18 percent of GDP) and accounted for about 30 percent of total government debt and for about 80 percent of domestic government domestic debt. Most external arrears outstanding at end-2006 (excluding those to Paris Club creditors) have been repaid, but new arrears—for a slightly lower amount—have been incurred in 2007; the authorities expect to clear all these arrears in by end-August 2008 (MEFP ¶18). In 2007 there has also been a large net repayment of domestic arrears, along the lines of a 10-year repayment plan that was adopted in 2002.

54. **Government debt amounted to $528 million, equivalent to 62 percent of GDP.** This included debt to external creditors, outstanding external and domestic arrears, and outstanding loans to banks and to the Port for $6.2 million that are gradually being repaid (Appendix Table 13).

55. **At present, public financing is only available from external official creditors.** Djibouti does not have access to international capital markets, and there is no market for domestic government debt. Under the currency board arrangement, the central bank is forbidden by law to lend to the government, and a domestic market for Treasury bills and bonds has not yet been established. Weaknesses in treasury management and cash planning have frequently resulted in the recourse to domestic and external payment arrears as a source of emergency short-term financing.

## B. Underlying Assumptions

56. **The medium-term macroeconomic framework has improved compared with the previous two DSAs, reflecting increased investors' interest in Djibouti and the expectation that agreement on a Fund-supported program would result in the implementation of policy measures aimed at enhancing competitiveness and sustaining growth (Appendix Box 4).** These measures, which would include a major fiscal adjustment and structural reforms, would enhance growth prospects, promote economic diversification and increase export capacity and foreign direct investment. The 2005 DSA assumed wide-ranging reforms but comparatively low FDI, and more moderate growth rates in the medium term. The 2007 DSA (CR/07/178) took into account the larger FDI flows, but was based instead on economic assumptions that did not incorporate comprehensive fiscal and structural reforms.

57. **Past historical trends do not provide a good basis for future projections because Djibouti's economy is undergoing a deep and rapid transformation led by foreign and public investment.** Favored by its strategic position at the mouth of the Red Sea and among major shipping routes, and taking advantage of the increased regional integration that followed the re-routing of most Ethiopian external trade through the Djibouti Port since 1998, Djibouti has been expanding its port and transportation infrastructure with the support of massive foreign direct investment from the Gulf, with the aim of transforming the country into a thriving regional trade and financial hub. At the same time, the authorities have pursued a program of public investment in social capital and infrastructure, aimed at reducing poverty and improving opportunities and access to basic services, with the support of external bilateral and multilateral donors. These investments are expected to result in a substantial acceleration of growth, which already started in 2006, and could be further enhanced by policy measures aimed at improving competitiveness.

58. **Prompted by foreign investment and productivity improvements under a Fund-supported program, real GDP is projected to achieve high and sustained growth rates in the medium term.** Large FDI inflows are expected between 2008 and 2013 in association with the completion of the new port, the building of an oil refinery and the establishment of new enterprises in the free-trade zone. Investment will then decline to more normal levels but will remain sustained over the next decade, enhanced by economic expansion and by the consolidation of Djibouti as a regional trading center. After 2018, FDI is projected to decline in percent of GDP and annual real GDP growth to slow down to a more sustainable steady state level. Imports will initially reflect foreign direct investment—that will prompt the purchase of equipment and other goods and services from

external suppliers—and will further increase in 2011 (together with exports) with the opening of the refinery. In the long term, productivity improvements and the expansion of domestic production will improve the balance of goods and services. The real exchange rate is expected to remain broadly stable, as upward pressures arising from capital inflows would be compensated by the mobilization of a largely underemployed labor force and by productivity improvements in the public administration and other nontradable sectors. This stability would facilitate the persistence of the fixed exchange peg to the U.S. dollar under the currency board arrangement.

### Appendix Box 4. Djibouti: Macroeconomic Assumptions

*The macroeconomic assumptions have improved considerably compared with the previous two DSAs, reflecting expectations of an agreement on a Fund-supported program. Growth and FDI inflows are projected to remain substantial over the next decade, prompting a marked growth in exports.*

**Real GDP growth** is projected to increase to 7 percent by 2009 and remain between 7–8 percent until 2018, to decline gradually thereafter to a more sustainable steady state rate of 3.5 percent. Persistent growth rates well above the historical average would stem from productivity improvements induced by structural reforms under a Fund-supported program as well as from the enhancement in physical and human capital resulting from the externally financed public investment program and from massive inflows of FDI (projected at an annual average of 16 percent of GDP until 2013 and 6 percent of GDP thereafter).

Average **inflation** is projected to peak at about 8 percent in 2008 and to stabilize at 3–3.5 percent by 2011. Fiscal restraint under a Fund-supported program and structural reforms aimed at increasing competitiveness in the local market, combined with the fixed dollar peg under the currency board arrangement, are expected to keep inflation in this small open economy close to the average world level.

The non-interest **current account deficit** is projected to widen to 3 percent of GDP in 2008 and 2009 and then decline to about 14 percent by 2013, reflecting the decline in FDI inflows after the completion of the new port and the refinery. The non-interest current account deficit is then projected to narrow more gradually to less than 6 percent by 2028. **Exports** are projected to increase threefold, from about 45 percent of GDP in 2008–10 to about 140 percent of GDP, with the opening of an oil refinery in 2011. This will also be reflected in an equivalent increase in **imports**. In the long term, as a result of economic diversification and the expansion of export-oriented and import-substitution activities, export growth is projected to outpace the growth of imports.

The **fiscal position** is projected to come into balance by 2011 as a result of fiscal adjustment under a Fund-supported program, and to remain balanced in the long term. The discipline enforced under the currency board arrangement, together with limited opportunities to borrow from domestic and external private sources, are projected to contain the overall deficit below 1 percent of GDP. Grant financing and the availability of concessional loans are projected to decline after 2013 as a result of the growth of the economy. It is assumed however that external loans to finance the public investment program will continue to be available from official creditors in the long term, albeit on less favorable terms.

Debt to **Paris Club creditors** that was originated before the cutoff date of March 31, 1998, is assumed to be rescheduled on classic non-concessional terms. The amounts eligible for rescheduling would include arrears outstanding at the start of the Fund-supported program as well as debt service payments coming due during the period of the program (September 2008–August 2011). According to the terms of reference agreed with the Paris Club in 2000, repayments would be made in ten equal semi-annual installments starting in 2013, and interest would be due at an appropriate market rate, that is here assumed at 5 percent. Similar treatment is also expected from non-Paris Club bilateral creditors.

59. **The substantial pick up in real GDP growth projected in the baseline scenario is a continuation of the trend seen in 2006 and 2007**, when growth started to accelerate driven by increases in FDI. Productive capacity would be expanded in the near future as several investment projects, including the new port and free zones (Emirati capital), are nearing completion, and other large projects, including a refinery financed by Kuwait, are also confirmed for the period 2008–13 (Table 8 in the staff report). These investments carried out in the context of strategic partnerships with large public and private foreign investors (such as U.A.E.'s Dubai Ports) are bringing about important structural changes in the economy. Dependence on imports will be substantially reduced with the desalinization plant and the refinery, and other investment projects (salt mining, cattle reexport facilities, port and logistics services, and new luxury hotels) will boost export capacity. Planned large public investments in infrastructure and social services will also help increase competitiveness in the long-term. These investments will be completed in the nearer term with initiatives to promote the development of a vibrant private sector with domestic SMEs dedicated to service the large foreign-owned projects. Supporting initiatives include the revised commerce and labor codes which will provide a stronger legal framework for doing business and investment, the reform of public utilities' enterprises, and prudent macroeconomic policies.

60. **This favorable growth scenario is expected to be underpinned by a prudent fiscal policy.** In the long term, external grants and revenue from external sources associated with the military bases are projected to decline in percent of GDP, but fiscal reforms and improvements in tax administration implemented under the Fund-supported program would enhance long-term revenue mobilization. A combination of wage restraint, a reorganization of the civil service, and improvements in expenditure management are expected to contain current expenditure. At the same, with financial support from external donors, the government would continue to implement a public investment program aimed at promoting social development and public infrastructure. It is also assumed that, in order to enhance its credibility and foster the confidence of domestic and foreign investors, the government would avoid the accumulation of new arrears. This will be facilitated by treasury reforms and improvements in cash management implemented under the program. All external arrears are projected to be repaid or rescheduled in 2008, and the repayment of domestic arrears is expected to continue broadly in line with the 10-year repayment schedule.

61. **The government is not expected to borrow from private sources in the foreseeable future.** Although negotiable government securities may be introduced for liquidity management purposes in the medium to long term, a full-fledged market for government debt is not expected to develop within the projection period. Economic expansion and policy credibility under a Fund-supported program could facilitate prospects for future access to international financial markets, but, in order to maintain macroeconomic stability and debt sustainability, the government is expected to continue to base its borrowing strategy on the mobilization of external financing from official creditors on favorable (if not fully concessional) terms. Economic growth is however expected to result in a lower availability of concessional financing, which will be reflected in a decline in the average grant element of new loans.

## C. Evaluation of External Debt Sustainability

62. **Under baseline assumptions, most external debt and debt service ratios are above the sustainability threshold, indicating a high risk of debt distress (Appendix Table 15a; Appendix Figure 4).** These ratios tend to improve in the medium and long-term, under the baseline scenario. The net present value (NPV) of debt, which at end-2007 stood at 50 percent of GDP, is projected to reach the 30 percent level considered sustainable for a country like Djibouti in 2016, mainly as a result of GDP growth, and remain close to that level thereafter. The NPV of debt-to-exports ratio is projected to cross the 100 percent threshold in 2009, and to improve considerably as a result of the opening of the oil refinery in 2011; even excluding this effect, this ratio is projected to maintain a downward trend, reaching about 40 percent by 2028 (Appendix Table 15a). Other debt sustainability indicators are well below their respective thresholds and are projected to improve in the medium term. The debt service-to-exports ratio remains, through the whole projection period, comfortably below the 15 percent threshold considered sustainable, even when exports of refined products are excluded from the denominator.

63. **However, external sustainability remains vulnerable to a worsening of external borrowing conditions and other external shocks.** If new borrowing could only be secured at interest rates 2 percentage points higher than under the baseline (A2 in Appendix Table 15b), the NPV of debt would remain above 50 percent of GDP throughout the projection period, with an upward trend after 2020. In the event of a high investment and low growth scenario (HI/LG, A3 in Appendix Table 15b), with new FDI failing to generate additional growth, annual real GDP growth would return to a steady state rate of around 3 percent by 2013 and the NPV of debt would reach 50 percent toward the end of the projection period. Djibouti could also be vulnerable to other types of external shocks whose effects have not been quantified, most notably less favorable inflows of FDI, a slower than projected increase in exports, and adverse developments in terms of trade and cross exchange rates. Lower FDI or lower exports would result in slower growth, weakening the improvement in debt sustainability produced by the endogenous debt dynamics (in the case of FDI, this effects could be partly compensated by a reduction in imports that could improve the current account balance); an unfavorable evolution of the terms of trade, resulting, for instance, from a protracted increase in international food and energy prices or by a devaluation of the U.S. dollar vis-à-vis other currencies, unless fully compensated by improvements in competitiveness (more likely to occur in the case of a devaluation of the U.S. dollar) would widen the current account deficit, increasing external borrowing needs. A devaluation of the dollar would also increase the domestic currency value of debt denominated in currencies other than the dollar (and not pegged to the dollar), which amounted at end-2007 to 85.4 percent of the total (excluding arrears) (Appendix Table 16).

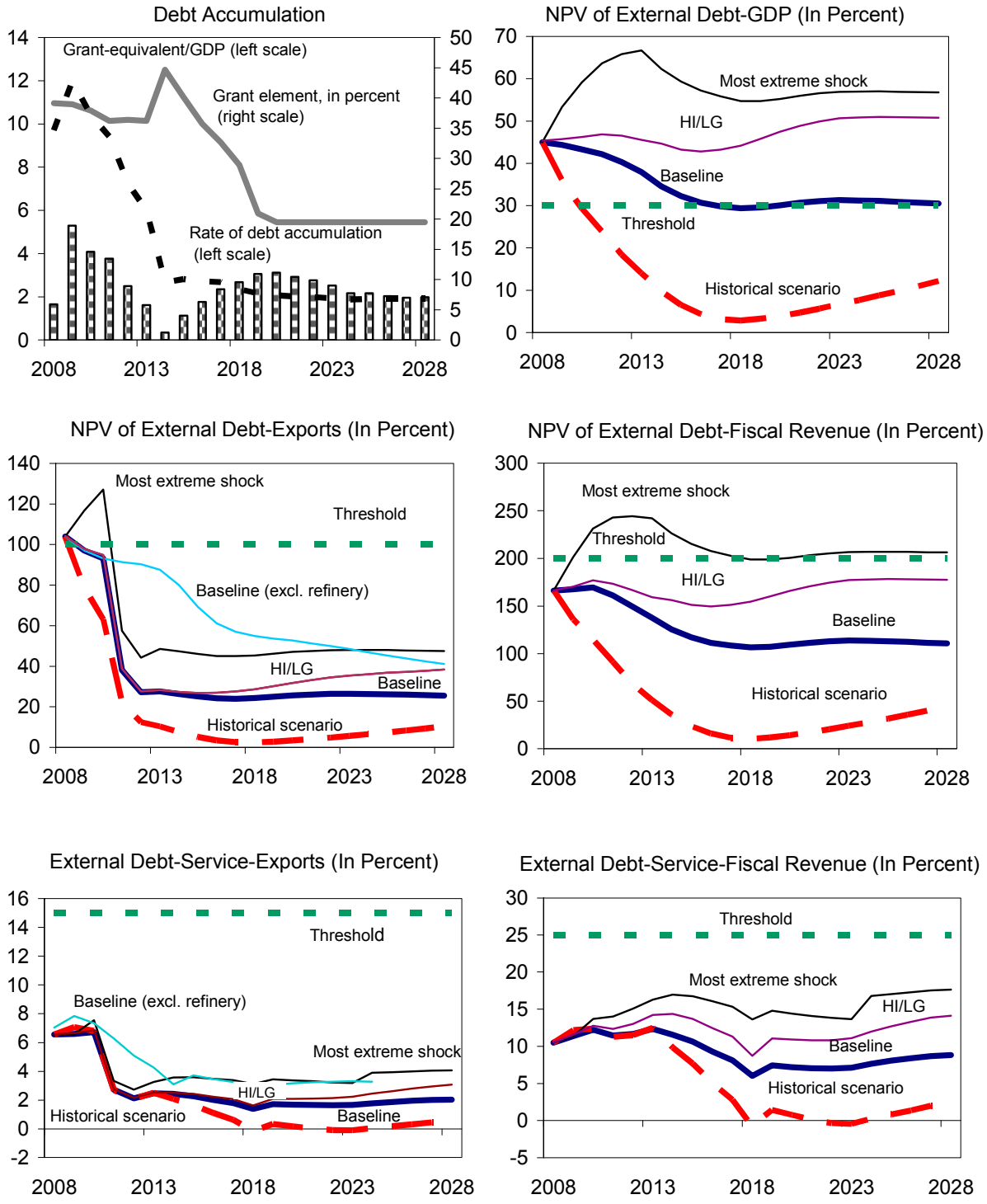
Appendix Table 15a. Djibouti: External Debt Sustainability Framework, Baseline Scenario, 2007–28 1/
(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average 2/ | Standard Deviation 2/ | Projections | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | | | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008-13 Average | 2018 | 2023 | 2028 | 2014-28 Average |
| **External debt (nominal) 1/** | **62.0** | **56.6** | **59.3** | | | **59.4** | **60.6** | **60.4** | **59.5** | **57.1** | **54.1** | | **41.8** | **40.4** | **37.3** | **41.5** |
| o/w public and publicly guaranteed (PPG) | 62.0 | 56.6 | 59.3 | | | 59.4 | 60.6 | 60.4 | 59.5 | 57.1 | 54.1 | | 41.8 | 40.4 | 37.3 | 41.5 |
| Change in external debt | -5.8 | -5.4 | 2.7 | | | 0.1 | 1.2 | -0.2 | -0.9 | -2.5 | -3.0 | | -1.3 | -0.4 | -0.5 | -1.1 |
| Identified net debt-creating flows | -9.2 | -11.4 | -3.6 | | | 6.2 | 8.8 | 6.4 | 4.1 | 2.9 | 0.9 | | 0.5 | 1.0 | 0.7 | 0.8 |
| **Non-interest current account deficit** | **2.4** | **13.9** | **24.0** | **5.7** | **8.5** | **32.7** | **31.8** | **26.4** | **22.1** | **16.1** | **12.8** | | **9.5** | **6.4** | **6.0** | **8.1** |
| Deficit in balance of goods and services | 14.5 | 23.0 | 32.6 | | | 40.6 | 39.4 | 33.1 | 28.6 | 21.6 | 18.1 | | 12.1 | 8.4 | 7.5 | 10.5 |
| Exports | 40.6 | 40.6 | 42.8 | | | 43.2 | 45.7 | 46.5 | 110.5 3/ | 148.7 | 137.5 | | 120.8 | 118.4 | 119.3 | 121.2 |
| Imports | 55.1 | 63.6 | 75.4 | | | 83.8 | 85.1 | 79.6 | 139.1 3/ | 170.3 | 155.6 | | 132.9 | 126.9 | 126.9 | 131.7 |
| Net current transfers (negative = inflow) | -1.8 | 2.7 | 3.2 | -1.4 | 2.7 | 1.8 | 1.5 | 1.9 | 1.8 | 1.7 | 1.6 | | 1.6 | 1.6 | 1.6 | 1.6 |
| o/w official | -3.9 | -2.4 | -1.3 | | | -2.3 | -2.2 | -1.5 | -1.4 | -1.3 | -1.2 | | -1.2 | -1.2 | -1.2 | -1.2 |
| Other current account flows (negative = net inflow) | -10.2 | -11.8 | -11.8 | | | -9.7 | -9.1 | -8.6 | -8.4 | -7.2 | -6.9 | | -4.2 | -3.6 | -3.2 | -4.1 |
| **Net FDI (negative = inflow)** | **-8.3** | **-21.3** | **-23.0** | **-7.0** | **9.0** | **-24.3** | **-20.5** | **-17.1** | **-14.7** | **-10.2** | **-9.2** | -16.0 | **-6.9** | **-4.8** | **-4.9** | **-6.0** |
| **Endogenous debt dynamics 4/** | **-3.3** | **-4.1** | **-4.6** | | | **-2.2** | **-2.6** | **-2.9** | **-3.2** | **-3.1** | **-2.7** | | **-2.1** | **-0.6** | **-0.4** | **-1.2** |
| Contribution from nominal interest rate | 0.8 | 0.8 | 0.8 | | | 0.9 | 1.0 | 1.1 | 1.0 | 1.0 | 1.0 | | 0.7 | 0.8 | 0.9 | 0.8 |
| Contribution from real GDP growth | -2.0 | -2.8 | -2.7 | | | -3.0 | -3.6 | -4.0 | -4.3 | -4.1 | -3.7 | | -2.7 | -1.4 | -1.3 | -2.0 |
| Contribution from price and exchange rate changes | -2.1 | -2.1 | -2.7 | | | … | … | … | … | … | … | | … | … | … | |
| **Residual 5/** | **3.4** | **6.0** | **6.3** | | | **-6.1** | **-7.6** | **-6.6** | **-5.0** | **-5.3** | **-3.9** | | **-1.8** | **-1.4** | **-1.3** | **-1.9** |
| o/w exceptional financing | -0.1 | -0.8 | -1.0 | | | -2.3 | -2.6 | -1.8 | -1.6 | -0.4 | 0.0 | | 0.0 | 0.0 | 0.0 | 0.0 |
| NPV of external debt 6/ | 57.3 | 52.8 | 49.8 | | | 44.9 | 44.3 | 43.3 | 42.2 | 40.3 | 37.9 | | 29.4 | 31.3 | 30.5 | 30.9 |
| In percent of exports | 141.1 | 129.9 | 116.4 | | | 103.9 | 97.1 | 93.1 | 38.2 | 27.1 | 27.6 | | 24.3 | 26.4 | 25.5 | 25.5 |
| **NPV of PPG external debt** | **57.3** | **52.8** | **49.8** | | | **44.9** | **44.3** | **43.3** | **42.2** | **40.3** | **37.9** | | **29.4** | **31.3** | **30.5** | **30.9** |
| **In percent of exports** | **141.1** | **129.9** | **116.4** | | | **103.9** | **97.1** | **93.1** | **38.2** | **27.1** | **27.6** | | **24.3** | **26.4** | **25.5** | **25.5** |
| **In percent of exports, excluding refined oil** | 141.1 | 129.9 | 116.4 | | | 103.9 | 97.1 | 93.1 | 91.4 | 90.3 | 87.6 | | 54.8 | 48.5 | 41.1 | 53.2 |
| **In percent of government revenues** | **185.5** | **169.8** | **165.1** | | | **166.3** | **167.5** | **169.4** | **161.0** | **149.7** | **137.8** | | **106.6** | **113.7** | **110.7** | **112.3** |
| **Debt service-to-exports ratio (in percent)** | **8.1** | **7.4** | **6.7** | | | **6.6** | **6.6** | **6.7** | **2.7** | **2.1** | **2.5** | | **1.4** | **1.7** | **2.0** | **1.9** |
| **Excluding exports of refined oil products** | 8.1 | 7.4 | 6.7 | | | 6.6 | 6.6 | 6.7 | 6.5 | 7.0 | 7.8 | | 3.1 | 3.0 | 3.3 | 3.9 |
| **PPG debt service-to-exports ratio (in percent)** | **8.1** | **7.4** | **6.7** | | | **6.6** | **6.6** | **6.7** | **2.7** | **2.1** | **2.5** | | **1.4** | **1.7** | **2.0** | **1.9** |
| **PPG debt service-to-revenue ratio (in percent)** | **10.7** | **9.6** | **9.5** | | | **10.5** | **11.4** | **12.2** | **11.5** | **11.7** | **12.3** | | **6.0** | **7.1** | **8.8** | **8.2** |
| Total gross financing need (billions of U.S. dollars) | 0.0 | 0.0 | 0.0 | | | 0.1 | 0.2 | 0.2 | 0.1 | 0.1 | 0.1 | | 0.1 | 0.1 | 0.2 | 0.1 |
| Non-interest current account deficit that stabilizes debt ratio | 8.3 | 19.3 | 21.4 | | | 32.6 | 30.7 | 26.6 | 22.9 | 18.6 | 15.8 | | 10.9 | 6.8 | 6.5 | 9.2 |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.2 | 4.8 | 5.3 | 3.1 | 1.4 | 5.9 | 6.9 | 7.3 | 7.9 | 7.7 | 7.2 | 7.1 | 7.0 | 3.7 | 3.6 | 5.0 |
| GDP deflator in US dollar terms (change in percent) | 3.1 | 3.5 | 5.0 | 2.6 | 1.2 | 8.1 | 6.0 | 4.2 | 3.5 | 3.0 | 3.0 | 4.6 | 3.5 | 3.5 | 3.5 | 3.5 |
| Effective interest rate (percent) 7/ | 1.2 | 1.4 | 1.5 | 1.4 | 0.1 | 1.7 | 2.0 | 2.0 | 1.9 | 1.9 | 2.0 | 1.9 | 1.7 | 2.2 | 2.5 | 2.0 |
| Growth of exports of G&S (US dollar terms, in percent) | 14.8 | 8.5 | 16.4 | 6.7 | 7.8 | 15.7 | 19.7 | 13.8 | 165.3 3/ | 49.3 | 2.1 | 44.3 | 7.6 | 7.7 | 7.3 | 7.7 |
| Growth of imports of G&S (US dollar terms, in percent) | 15.7 | 25.2 | 31.1 | 10.6 | 13.5 | 27.3 | 15.0 | 4.6 | 95.1 3/ | 35.8 | 0.9 | 29.8 | 7.0 | 7.2 | 7.2 | 7.2 |
| Grant element of new public sector borrowing (in percent) | … | … | … | … | … | 39.1 | 39.0 | 37.9 | 36.2 | 36.4 | 36.3 | 37.5 | 28.9 | 19.5 | 19.5 | 25.2 |
| Aid flows (in millions of US dollars) 8/ | 83.1 | 68.8 | 100.7 | | | 147.6 | 186.1 | 178.0 | 181.4 | 166.7 | 163.5 | | 54.6 | 51.3 | 72.6 | 61.2 |
| o/w Grants | 43.8 | 29.7 | 42.0 | | | 57.8 | 91.5 | 91.9 | 94.8 | 79.8 | 72.6 | | 34.6 | 51.3 | 72.6 | 45.5 |
| o/w Concessional loans | 28.5 | 40.8 | 51.9 | | | 87.4 | 99.0 | 91.1 | 90.9 | 83.7 | 80.1 | | 18.0 | 0.0 | 0.0 | 7.1 |
| Grant-equivalent financing (in percent of GDP) 9/ | … | … | … | | | 9.7 | 12.0 | 10.4 | 9.4 | 7.2 | 6.0 | | 2.3 | 1.9 | 1.9 | 2.2 |
| Grant-equivalent financing (in percent of external financing) 9/ | … | … | … | | | 62.3 | 67.6 | 68.3 | 67.9 | 67.5 | 66.6 | | 46.2 | 40.2 | 40.3 | 43.7 |
| *Memorandum items:* | | | | | | | | | | | | | | | | |
| Nominal GDP (millions of US dollars) | 708.8 | 768.9 | 849.8 | | | 972.8 | 1101.9 | 1232.0 | 1375.6 | 1525.6 | 1684.7 | | 2806.4 | 4157.2 | 5886.8 | **3691** |
| (NPVt-NPVt-1)/GDPt-1 (in percent) | | | | | | 1.6 | **5.3** | 4.1 | 3.8 | 2.5 | 1.6 | 3.2 | 2.7 | 2.5 | 2.0 | 2.2 |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ The sharp change in imports and exports reflects the start of oil refining.
4/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
5/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
6/ Assumes that NPV of private sector debt is equivalent to its face value.
7/ Current-year interest payments divided by previous period debt stock.
8/ Defined as grants, concessional loans, and debt relief.
9/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Appendix Figure 4. Djibouti: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2008–28




Source: Staff projections and simulations.

Appendix Table 15b. Djibouti: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007–28
(In percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2018 | 2028 |
| **NPV of debt-to-GDP ratio** | | | | | | | | |
| **Baseline** | 45 | 44 | 43 | 42 | 40 | 38 | 29 | 30 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008-27 1/ | 45 | 36 | 29 | 24 | 18 | 14 | 3 | 12 |
| A2. New public sector loans on less favorable terms in 2008-27 2/ | 45 | 53 | 59 | 64 | 66 | 67 | 55 | 57 |
| A3. High FDI does not contribute to higher growth in 2008-27 | 45 | 46 | 46 | 47 | 47 | 46 | 44 | 51 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2008-09 | 45 | 47 | 48 | 47 | 45 | 42 | 33 | 34 |
| B2. Export value growth at historical average minus one standard deviation in 2008-09 3/ | 45 | 50 | 59 | 57 | 54 | 51 | 37 | 33 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09 | 45 | 46 | 47 | 45 | 43 | 41 | 32 | 33 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/ | 45 | 61 | 73 | 70 | 67 | 63 | 44 | 35 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 45 | 52 | 62 | 60 | 57 | 54 | 40 | 37 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/ | 45 | 62 | 60 | 59 | 56 | 53 | 41 | 43 |
| **NPV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 104 | 97 | 93 | 38 | 27 | 28 | 24 | 26 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2007-26 1/ | 104 | 79 | 63 | 22 | 12 | 10 | 2 | 10 |
| A2. New public sector loans on less favorable terms in 2007-26 2/ | 104 | 117 | 127 | 58 | 44 | 48 | 45 | 48 |
| A3. High FDI does not contribute to higher growth in 2008-27 | 104 | 98 | 95 | 39 | 28 | 28 | 29 | 38 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2008-09 | 104 | 97 | 93 | 38 | 27 | 28 | 24 | 26 |
| B2. Export value growth at historical average minus one standard deviation in 2008-09 3/ | 104 | 134 | 177 | 72 | 51 | 52 | 43 | 38 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09 | 104 | 97 | 93 | 38 | 27 | 28 | 24 | 26 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/ | 104 | 134 | 158 | 63 | 45 | 46 | 36 | 29 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 104 | 123 | 148 | 60 | 43 | 43 | 36 | 35 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/ | 104 | 97 | 93 | 38 | 27 | 28 | 24 | 26 |
| **NPV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 166 | 167 | 169 | 161 | 150 | 138 | 107 | 111 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2007-26 1/ | 166 | 136 | 115 | 91 | 69 | 51 | 10 | 44 |
| A2. New public sector loans on less favorable terms in 2007-26 2/ | 166 | 201 | 231 | 243 | 244 | 242 | 199 | 206 |
| A3. High FDI does not contribute to higher growth in 2008-27 | 166 | 170 | 177 | 173 | 167 | 159 | 154 | 177 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2008-09 | 166 | 176 | 188 | 179 | 166 | 153 | 118 | 123 |
| B2. Export value growth at historical average minus one standard deviation in 2008-09 3/ | 166 | 190 | 231 | 217 | 201 | 185 | 134 | 119 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09 | 166 | 175 | 182 | 173 | 161 | 148 | 115 | 119 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/ | 166 | 232 | 287 | 268 | 247 | 228 | 158 | 127 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 166 | 197 | 242 | 228 | 212 | 195 | 144 | 135 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/ | 166 | 234 | 236 | 225 | 209 | 192 | 149 | 154 |
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 7 | 7 | 7 | 3 | 2 | 2 | 1 | 2 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008-27 1/ | 7 | 7 | 7 | 3 | 2 | 2 | 0 | 1 |
| A2. New public sector loans on less favorable terms in 2008-27 2/ | 7 | 7 | 8 | 3 | 3 | 3 | 3 | 4 |
| A3. High FDI does not contribute to higher growth in 2008-27 | 7 | 7 | 7 | 3 | 2 | 3 | 2 | 3 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2008-09 | 7 | 7 | 7 | 3 | 2 | 2 | 1 | 2 |
| B2. Export value growth at historical average minus one standard deviation in 2008-09 3/ | 7 | 8 | 10 | 4 | 3 | 4 | 3 | 3 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09 | 7 | 7 | 7 | 3 | 2 | 2 | 1 | 2 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/ | 7 | 7 | 8 | 4 | 3 | 3 | 2 | 3 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 7 | 8 | 9 | 4 | 3 | 3 | 2 | 3 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/ | 7 | 7 | 7 | 3 | 2 | 2 | 1 | 2 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 11 | 11 | 12 | 11 | 12 | 12 | 6 | 9 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2008-27 1/ | 11 | 12 | 12 | 11 | 12 | 12 | -1 | 2 |
| A2. New public sector loans on less favorable terms in 2008-27 2/ | 11 | 11 | 14 | 14 | 15 | 16 | 14 | 18 |
| A3. High FDI does not contribute to higher growth in 2008-27 | 11 | 12 | 13 | 12 | 13 | 14 | 9 | 14 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2008-09 | 11 | 12 | 14 | 13 | 13 | 14 | 7 | 10 |
| B2. Export value growth at historical average minus one standard deviation in 2008-09 3/ | 11 | 11 | 13 | 13 | 13 | 14 | 9 | 10 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09 | 11 | 12 | 13 | 12 | 13 | 13 | 6 | 9 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/ | 11 | 11 | 14 | 15 | 15 | 15 | 11 | 11 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 11 | 12 | 15 | 15 | 15 | 15 | 9 | 11 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/ | 11 | 16 | 17 | 16 | 16 | 17 | 8 | 12 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24 |

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicit an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Appendix Table 16. Djibouti: Composition by Currency of External Debt Outstanding at End-2007 (Excluding Arrears)

| Currency | Amount of Debt in U.S. dollars (millions) | Percent of Total External Debt |
|---|---|---|
| SDR and SDR-related currency units | 287.7 | 60.9 |
| U.S. dollars and currencies pegged to the dollar | 68.6 | 14.6 |
| Euro | 32.5 | 6.9 |
| Other currencies | 82.3 | 17.5 |
| **Total** | **469.1** | **100.0** |

## D. Evaluation of Fiscal Sustainability

64. **Fiscal sustainability indicators are also projected to improve remarkably in the medium term (Appendix Table 17a; Appendix Figure 5).** While there are no defined thresholds for government debt, the NPV of government debt is projected to decline from 55 percent of GDP (and about 160 percent of fiscal revenue and grants) in 2007 to about 33 percent of GDP (and about 100 percent of fiscal revenue and grants) by 2011, as a result of the fiscal adjustment that would be implemented under a Fund-supported program, including a large repayment of arrears. These ratios are projected to improve further in the long term as sustained growth, improvements in revenue collection, and a more efficient expenditure execution contain the primary deficit well below the estimated debt-stabilizing value. These improvements would occur under the assumption of a gradual decline in external revenue from the French and U.S. military bases, and of sustained rates of public investment. The debt service-fiscal revenue ratio is projected to remain at a comparatively low level, at 3–5 percent, throughout the projection period.

65. **A temporary setback in growth in the short term could result in a much less favorable evolution of fiscal sustainability .** If real GDP growth in 2009 and 2010 turned out close to 1.6 percent, equivalent to its historical average less one standard deviation (B1 in Appendix Table 17b), without further adjustments in public expenditure, the primary deficit would widen to a long-term average of about 3 percent, which would result in a renewed accumulation of debt after the initial decline related to the repayment of arrears. The NPV of debt would eventually climb back to about 50 percent of GDP and 200 percent of revenue by 2028, while the debt service would increase to absorb about 20 percent of fiscal revenue by 2028, raising concerns about debt sustainability.

Appendix Table 17a. Djibouti: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005–28

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Estimate | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | Historical Average 5/ | Standard Deviation 5/ | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008-13 Average | 2018 | 2023 | 2028 | 2014-28 Average |
| **Public sector debt 1/** | 67.6 | 63.2 | 62.1 | | | 53.4 | 49.1 | 45.6 | 41.0 | 36.9 | 34.1 | 43.4 | 22.3 | 17.9 | 16.0 | 20.7 |
| o/w foreign-currency denominated | 40.3 | 38.0 | 40.1 | | | 35.9 | 35.6 | 34.9 | 32.5 | 30.2 | 28.4 | 32.9 | 20.9 | 17.7 | 15.9 | 19.5 |
| Change in public sector debt | -6.1 | -4.4 | -1.1 | | | -8.7 | -4.3 | -3.5 | -4.6 | -4.0 | -2.8 | -4.7 | -1.8 | -0.3 | -0.3 | -1.2 |
| Identified debt-creating flows | -4.7 | -1.6 | -3.4 | | | -5.9 | -4.4 | -3.5 | -4.7 | -3.8 | -3.2 | -4.3 | -1.8 | -0.3 | -0.3 | -1.2 |
| Primary deficit | -0.7 | 2.0 | 2.2 | 1.6 | 1.1 | 1.5 | 1.3 | 1.1 | -0.5 | -0.3 | -0.3 | 0.5 | 0.1 | 0.6 | 0.4 | 0.1 |
| Revenue and grants | 37.1 | 34.9 | 35.1 | | | 32.9 | 34.8 | 33.0 | 33.1 | 32.1 | 31.8 | 33.0 | 28.1 | 26.7 | 26.2 | 27.8 |
| *of which* : grants | 6.2 | 3.9 | 4.9 | | | 5.9 | 8.3 | 7.5 | 6.9 | 5.2 | 4.3 | 6.4 | 1.6 | 1.2 | 1.2 | 1.7 |
| Primary (noninterest) expenditure | 36.4 | 37.0 | 37.3 | | | 34.4 | 36.1 | 34.1 | 32.6 | 31.8 | 31.6 | 33.4 | 28.2 | 27.3 | 26.6 | 27.9 |
| Automatic debt dynamics | -4.0 | -4.8 | -5.6 | | | -7.4 | -5.7 | -4.6 | -4.2 | -3.5 | -3.0 | -4.7 | -1.9 | -0.9 | -0.7 | -1.3 |
| Contribution from interest rate/growth differential | -4.0 | -4.8 | -5.6 | | | -7.4 | -5.7 | -4.6 | -4.2 | -3.5 | -3.0 | -4.7 | -1.9 | -0.9 | -0.7 | -1.3 |
| *of which* : contribution from average real interest rate | -1.7 | -1.7 | -2.4 | | | -4.0 | -2.3 | -1.3 | -0.9 | -0.6 | -0.5 | -1.6 | -0.3 | -0.3 | -0.2 | -0.2 |
| *of which* : contribution from real GDP growth | -2.3 | -3.1 | -3.2 | | | -3.4 | -3.4 | -3.3 | -3.3 | -2.9 | -2.5 | -3.2 | -1.6 | -0.7 | -0.6 | -1.1 |
| Contribution from real exchange rate depreciation | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | ... | ... | ... | |
| Other identified debt-creating flows | 0.0 | 1.2 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Other | 0.0 | 1.2 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Residual, including asset changes | -1.4 | -2.8 | 2.4 | | | -2.8 | 0.1 | 0.0 | 0.0 | -0.2 | 0.4 | -0.4 | 0.0 | 0.0 | 0.0 | 0.0 |
| **NPV of public sector debt** | 63.1 | 58.1 | 55.0 | | | 45.6 | 40.9 | 37.2 | 33.2 | 29.6 | 26.7 | 35.5 | 15.7 | 13.5 | 12.8 | 15.4 |
| o/w foreign-currency denominated | 35.8 | 33.0 | 32.9 | | | 28.2 | 27.4 | 26.5 | 24.8 | 22.9 | 21.0 | 25.1 | 14.3 | 13.2 | 12.6 | 14.2 |
| o/w external | ... | ... | 32.9 | | | 28.2 | 27.4 | 26.5 | 24.8 | 22.9 | 21.0 | 25.1 | 14.3 | 13.2 | 12.6 | 14.2 |
| NPV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Gross financing need 2/ | 0.4 | 3.3 | 3.4 | | | 3.0 | 2.5 | 2.5 | 1.0 | 1.2 | 1.7 | 2.0 | 1.0 | 1.4 | 1.4 | 1.2 |
| NPV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 156.7 | | | 138.5 | 117.6 | 112.6 | 100.3 | 92.1 | 83.7 | 107.5 | 56.0 | 50.3 | 48.9 | 54.9 |
| NPV of public sector debt-to-revenue ratio (in percent) | ... | ... | 182.4 | | | 169.0 | 154.5 | 145.5 | 126.7 | 110.0 | 96.8 | 133.8 | 59.4 | 52.8 | 51.3 | 58.6 |
| o/w external 3/ | ... | ... | 109.3 | | | 104.4 | 103.4 | 103.7 | 94.5 | 85.1 | 76.3 | 94.6 | 54.0 | 51.8 | 50.6 | 54.2 |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 2.9 | 3.7 | 3.6 | | | 4.5 | 3.5 | 4.3 | 4.3 | 4.8 | 6.1 | 4.6 | 3.1 | 3.2 | 3.8 | 4.0 |
| Debt service-to-revenue ratio (in percent) 4/ | 3.4 | 4.2 | 4.2 | | | 5.5 | 4.6 | 5.6 | 5.5 | 5.7 | 7.1 | 5.7 | 3.3 | 3.3 | 4.0 | 4.2 |
| Primary deficit that stabilizes the debt-to-GDP ratio | 5.4 | 6.4 | 3.2 | | | 10.2 | 5.6 | 4.6 | 4.2 | 3.7 | 2.6 | 5.1 | 1.9 | 0.9 | 0.7 | 1.3 |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.2 | 4.8 | 5.3 | 3.1 | 1.4 | 5.9 | 6.9 | 7.3 | 7.9 | 7.7 | 7.2 | 7.1 | 7.0 | 3.7 | 3.6 | 5.0 |
| Average nominal interest rate on forex debt (in percent) | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.0 | 0.0 | 0.1 | 2.0 | 2.0 | 2.4 | 2.6 |
| Average real interest rate on domestic currency debt (in percent) | -1.7 | -1.8 | -3.2 | -1.4 | 1.0 | -5.6 | -2.7 | 0.4 | 2.1 | 3.8 | 5.3 | | ... | ... | ... | ... |
| Real exchange rate depreciation (in percent, + indicates depreciation) | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 3.1 | 3.5 | 5.0 | 2.6 | 1.2 | 8.1 | 6.0 | 4.2 | 3.5 | 3.0 | 3.0 | 4.6 | 3.5 | 3.5 | 3.5 | 3.5 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 1.1 | 6.4 | 6.1 | 4.5 | 7.1 | -2.1 | 12.0 | 1.4 | 3.1 | 5.1 | 6.4 | 4.3 | 5.6 | 2.8 | 3.3 | 3.8 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 28.2 | 19.5 | 19.5 | 24.5 |

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

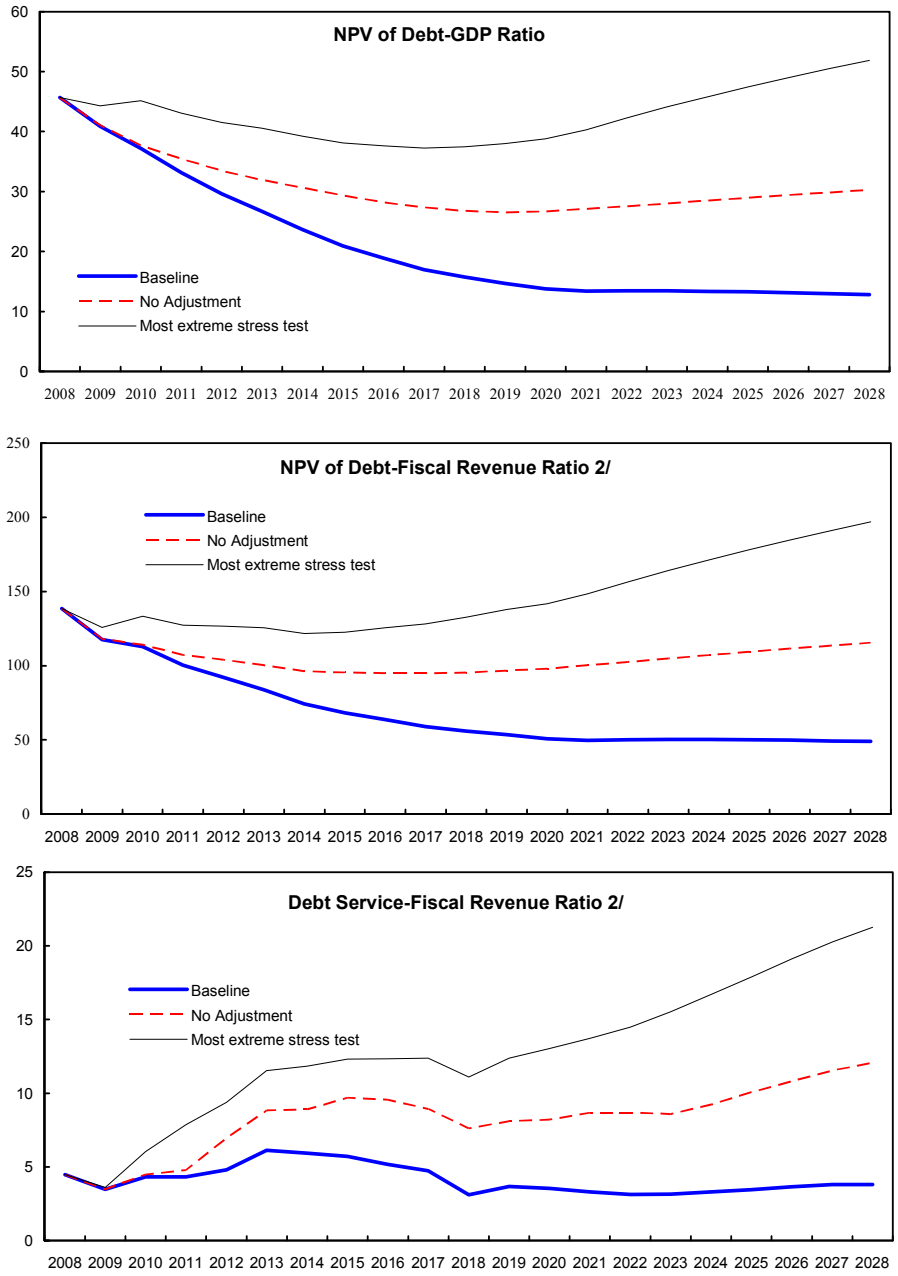
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

**Appendix Figure 5. Djibouti: Indicators of Public Debt Under Alternative Scenarios, 2008–28 1/**




Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2018.
2/ Revenue including grants.

Appendix Table 17b. Djibouti: Sensitivity Analysis for Key Indicators of Public Debt, 2008–28

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2018 | 2028 |
| **NPV of Debt-GDP Ratio** | | | | | | | | |
| **Baseline** | 46 | 41 | 37 | 33 | 30 | 27 | 16 | 13 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 46 | 43 | 41 | 40 | 39 | 39 | 37 | 41 |
| A2. Primary balance is unchanged from 2008 | 46 | 41 | 38 | 35 | 33 | 32 | 27 | 30 |
| A3. Permanently lower GDP growth 1/ | 46 | 41 | 38 | 34 | 31 | 28 | 19 | 26 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 46 | 44 | 45 | 43 | 41 | 41 | 37 | 52 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 46 | 42 | 40 | 36 | 32 | 29 | 17 | 14 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 46 | 43 | 42 | 37 | 33 | 29 | 16 | 12 |
| B4. One-time 30 percent real depreciation in 2009 | 46 | 51 | 46 | 41 | 36 | 32 | 19 | 16 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 46 | 50 | 46 | 41 | 37 | 34 | 21 | 16 |
| **NPV of Debt-Fiscal Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 139 | 118 | 113 | 100 | 92 | 84 | 56 | 49 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 138 | 121 | 121 | 118 | 119 | 118 | 130 | 152 |
| A2. Primary balance is unchanged from 2008 | 138 | 118 | 114 | 107 | 104 | 100 | 95 | 115 |
| A3. Permanently lower GDP growth 1/ | 139 | 118 | 114 | 102 | 95 | 88 | 69 | 100 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 139 | 126 | 133 | 127 | 127 | 125 | 133 | 197 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 139 | 121 | 121 | 108 | 99 | 91 | 61 | 53 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 139 | 124 | 125 | 110 | 101 | 91 | 57 | 46 |
| B4. One-time 30 percent real depreciation in 2009 | 139 | 147 | 139 | 123 | 113 | 102 | 69 | 61 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 139 | 144 | 139 | 125 | 116 | 106 | 74 | 63 |
| **Debt Service-Fiscal Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 4 | 3 | 4 | 4 | 5 | 6 | 3 | 4 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 4 | 4 | 5 | 5 | 8 | 10 | 10 | 16 |
| A2. Primary balance is unchanged from 2008 | 4 | 3 | 4 | 5 | 7 | 9 | 8 | 12 |
| A3. Permanently lower GDP growth 1/ | 4 | 3 | 4 | 5 | 5 | 7 | 5 | 10 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 4 | 4 | 6 | 8 | 9 | 12 | 11 | 21 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 4 | 3 | 6 | 6 | 6 | 7 | 3 | 5 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 4 | 4 | 5 | 6 | 5 | 6 | 3 | 3 |
| B4. One-time 30 percent real depreciation in 2009 | 4 | 4 | 4 | 4 | 5 | 6 | 4 | 5 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 4 | 3 | 14 | 8 | 7 | 8 | 4 | 6 |

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.